



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Lakic

Product name: kasugamycin 2% W/V SL

Product use: Fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a fungicide to control fungal and bacterial diseases affecting rice, vegetables and fruit. On rice, it controls diseases caused by *Pyricularia oryzae* and *Pseudomonas glumae* (bacterial grain rot), with ground and aerial applications and seedling diseases caused by various bacterial pathogens. In sugar beet, it controls *Cercospora beticola*. Also used to control plant diseases in various crops: *Pseudomonas syringae* pv. *lachrymans* in cucumbers; *Colletotrichum lagenarium* in melons and water melons; *Cladosporium fulvum* and *Corynebacterium michiganense* in tomatoes; *Mycovellosiella* spp. in aubergines; *Cercospora* spp. in celery; *Erwinia carotovora* in potatoes, carrots and onions; *Pseudomonas* spp. (halo blight) in kidney beans; *Xanthomonas* spp. in paprika; *Venturia* sp. in apples; *Pseudomonas* spp. in kiwifruit; etc.

2. HAZARD IDENTIFICATION

Hazard classification:

Likely Routes of Exposure: Oral, skin contact and inhalation

Potential Health Hazard: May cause slightly eye irritation

Potential Environmental Hazard: N/A

Explosion hazard: No hazard.

Warning Statements:

Keep out of reach of children.



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3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 3-O-[2-amino-4-[(carboxyiminomethyl) amino]-2,3,4,6 tetradeoxy-"-D-arabino-hexopyranosyl]-D-chiro-inositol

Common name: kasugamycin

Chemical class: Aminoglycoside Antibiotic Fungicide

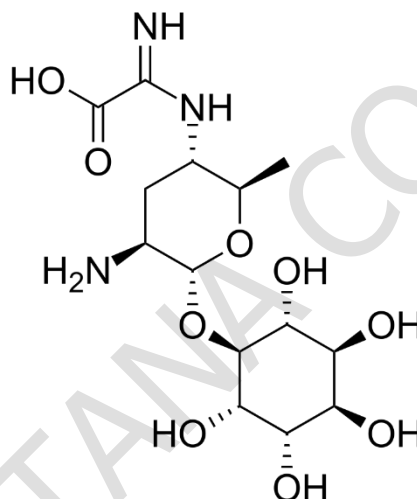
CAS Number: 6980-18-3

Ingredients and composition: Active ingredient – kasugamycin 2% W/V

Formula: $C_{14}H_{25}N_3O_9$

Molecular weight: 343.2 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

Eye: Rinse out with plenty of water with the eyelid held wide open for at least 15 minutes and seek medical attention.

Skin: Remove contaminated clothing and wash the affected area with water followed with soap and water.

Inhaled: Immediately remove from exposure to fresh air and seek medical attention if necessary.

Ingestion: If swallowed seek medical advice immediately and show the label if possible. Do not induce vomiting. Wash out mouth with water. Keep patient at rest and seek medical attention immediately.

Note to medical personnel:

There is no special antidote, treat according to symptom.



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5. FIRE FIGHTING MEASURES

Extinguishing Media:

Water spray, carbon dioxide, foam or dry powder.

Hazardous Combustion Products:

Carbon monoxide, carbon dioxide, nitrogen oxides.

Fire Fighting Equipment:

Wear self-contained breathing apparatus and protective clothing (chemical protection suit, boots and plastic or rubber gloves). Do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions

Ensure suitable personal protection during removal of spillages.

Environmental Precautions

Keep the area isolated and do not allow contamination to enter the sewerage system.

If the product has entered the sewerage system or contaminated a public place advise the appropriate authority.

Methods for cleaning up

Contain/absorb the spillage with a suitable inert absorbent material which should then be transferred to secure containers and sent for disposal. Wash the spillage area with water. Washings must be prevented from entering surface water drains.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Safe Handling:

Keep product in original tightly closed container; do not get on skin, eyes or clothing; do not eat drink or smoke when handling; wash hands before eating drinking or smoking. Provide good ventilation of working area.

Storage Conditions:

Store in a dry area in the original containers, access should only be for authorized persons. Keep out of reach of children. Keep away from food, drink and animal feeding stuffs. Protect from frost. Keep away from direct sunlight.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering measures to reduce exposure

Ensure adequate ventilation especially in confined areas.



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Threshold Limit Values (TLVs): N/A

Determination Method: HPLC

Engineering Control - Personal Protective Equipment

Eyes Protection:

Chemical safety glasses or goggles. Have eye-washing facilities readily available where eye contact can occur.

Skin Protection:

Wear chemical resistant gloves.

Respiratory Protection:

Wear appropriate respirator.

Industrial Hygiene

Immediately change contaminated clothing, dispose of if decontamination is not feasible. Keep away from food, drink and animal feeding stuffs. Wash hands before breaks and immediately after handling the product. When using, do not eat, drink or smoke.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

These physical data are typical values based on material tested but may vary from sample to sample. Typical values should not be construed as a guaranteed analysis of any specific lot or as specifications for the product.

Physical stage:	Liquid
Color:	Green
Odor:	N/A
Melting Point:	N/A
Flash Point:	No flash point
Autoflammability:	Not auto-flammable
Vapor Pressure:	7.2×10^{-4} mPa (20 °C)
pH:	3.0-5.0
Bulk density:	1.0160g/cm ³

10. STABILITY AND REACTIVITY

Stability

Stable at acid and neutral ambient conditions.

Hazardous decomposition

Carbon monoxide, carbon dioxide, nitrogen oxide, nitrogen dioxide. If product is heated above decomposition temperature, toxic vapors will be released.



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Materials to avoid/Reactivity

Avoid strong alkaline pesticide.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Kasugamycin:

LD50 (Oral, rat):	> 5,000 mg/kg body weight
LD50 (Dermal, rat):	> 2,000 mg/kg body weight
LC50 (Inhalation, rat, 4h):	2,000 mg/L
Skin irritation (rabbit):	Not irritant.
Eye irritation (rabbit):	Slightly irritation to rabbit
Skin sensitization:	Not a sensitizer to White guinea pig

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, Kasugamycin:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h):	> 100 mg/L
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Invertebrates – Acute toxicity:

Water flea (Daphnia magna),

EC50 (48 h):	65 mg/L
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Invertebrates – Acute toxicity:

Algae (Selenastrum capricornutum),

EC50:	12 mg/L
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Birds – Toxicity:

Bobwhite quail,

LD50:	> 4,000 mg/kg
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Arthropod – Toxicity:

Honey bee (Apis mellifera L.),

LD50 (oral):	> 100 ug/bee
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13. DISPOSAL CONSIDERATIONS

Waste from residues/unused products:

In accordance with local and national regulations. Do not allow material to contaminate ground water system. Do not contaminate surface water.

Container:

Wash out container thoroughly, empty washings into spray tank and dispose of safely.

Do not contaminate ponds, waterways or ditches with chemical or used container. Do not re-use empty containers. Burn empty bags or bury in a suitable location at a landfill.

14. TRANSPORT INFORMATION

UN proper shipping name: N/A

UN Number: N/A

Transport hazard class: N/A

Packaging group: N/A

15. REGULATORY INFORMATION

This safety datasheet complies with the requirements of Regulation on the Safe Management of Hazardous Chemicals in China.

16. OTHER INFORMATION

This SDS completes technical use sheets but does not replace them. Information herein is based on our present knowledge concerning the product, at the edition date. It is honestly given. The attention of users is drawn to potential risks taken when the product is used for other uses than those for which it is made. The user has to know and comply with all regulations concerning its activity. It remains the user's own responsibility to make sure that the information is appropriate and complete for his special use of this product. The aim of all the regulations mentioned is to help the person concerned to comply with the rules which are his own responsibility. This listing cannot be considered as exhaustive. The person concerned has to make sure that he has no other obligations due to texts specific to particular applications.